Causal Models

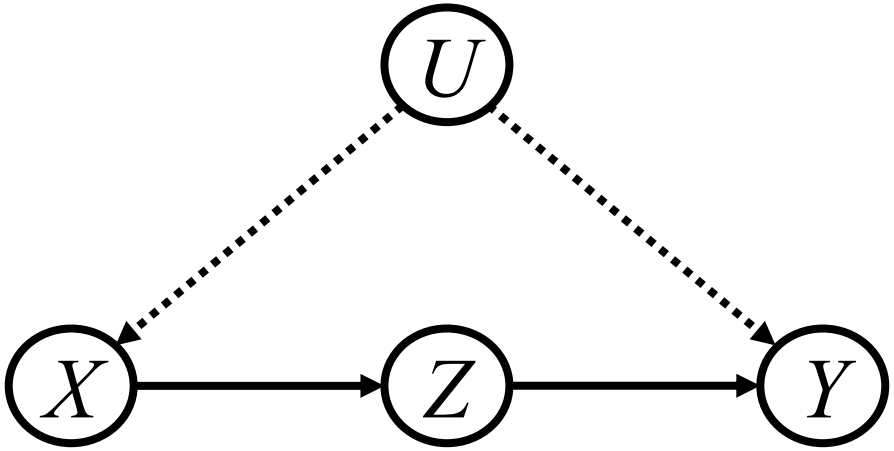
--Frontdoor Adjustment

Outline

- Frontdoor adjustment
- Representative works

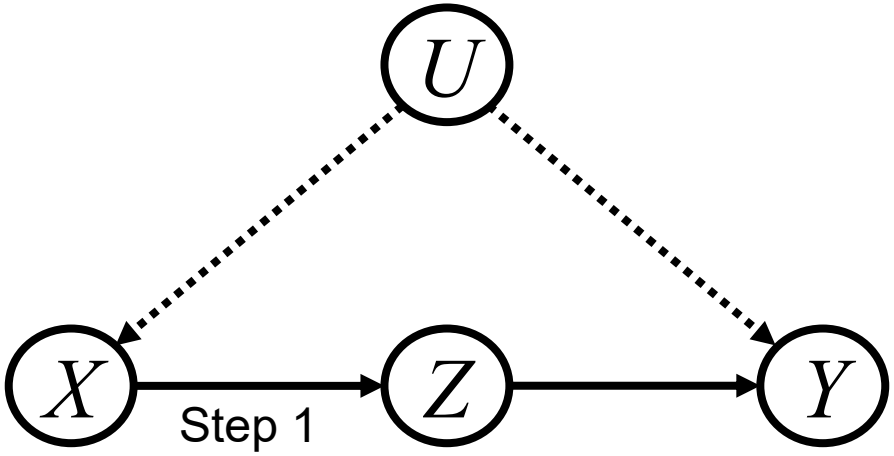
Frontdoor Adjustment: Goal

- Goal: Identify the causal effect of X on Y



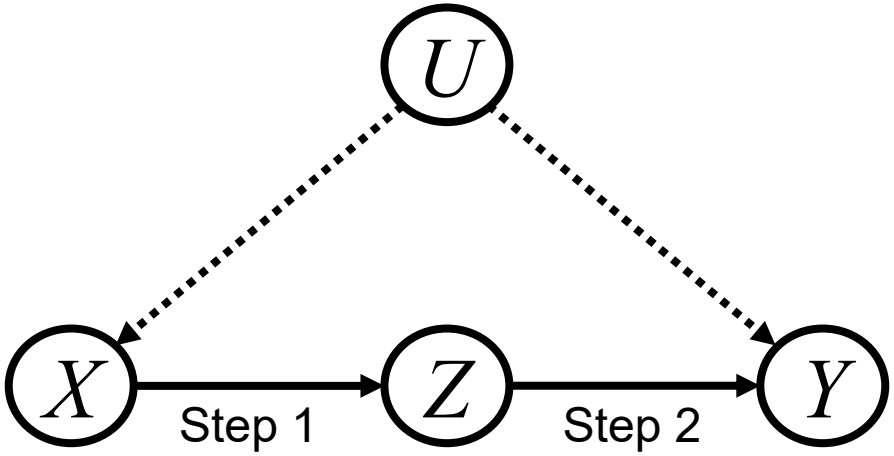
Frontdoor Adjustment: Big Picture

- Step 1: Identify the causal effect of X on Z



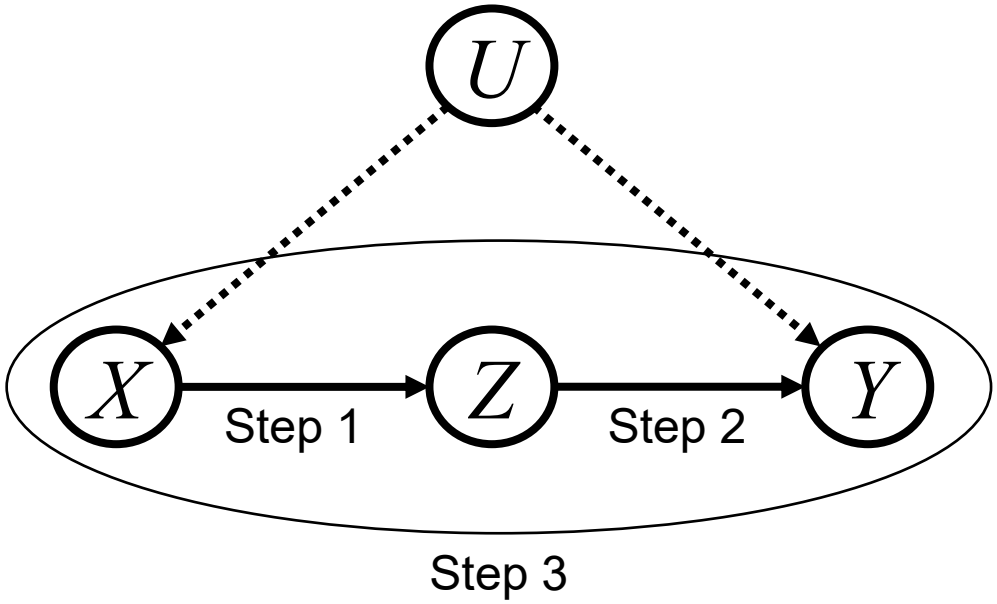
Frontdoor Adjustment: Big Picture

- Step 1: Identify the causal effect of X on Z
- Step 2: Identify the causal effect of Z on Y



Frontdoor Adjustment: Big Picture

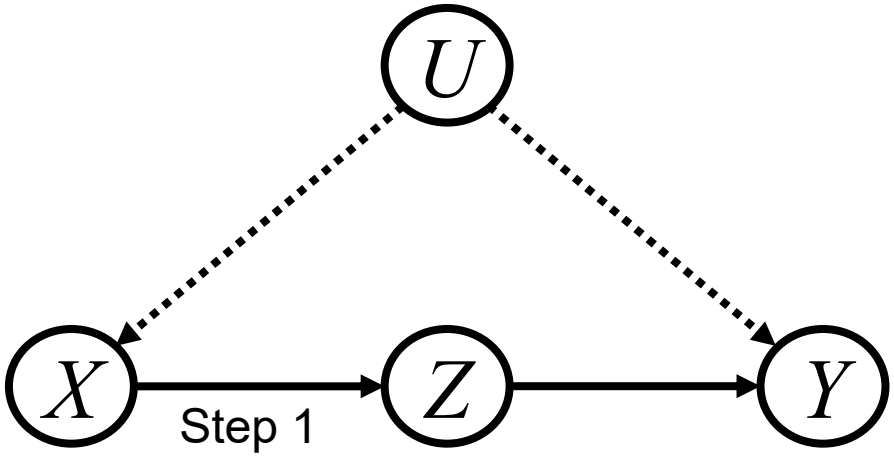
- Step 1: Identify the causal effect of X on Z
- Step 2: Identify the causal effect of Z on Y
- Step 3: Combine the above steps to identify the causal effect of X on Y



Frontdoor Adjustment: Step 1

- Identify the causal effect of *X* on *Z*

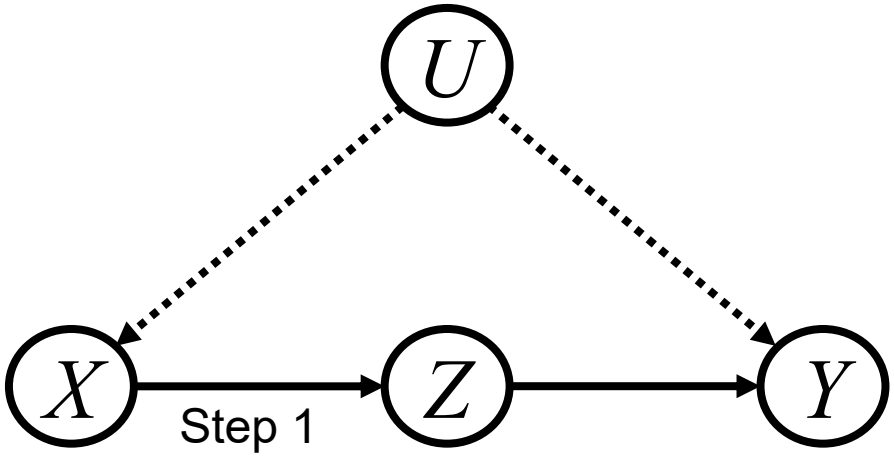
$$P(z|do(x))$$



Frontdoor Adjustment: Step 1

- Identify the causal effect of *X* on *Z*

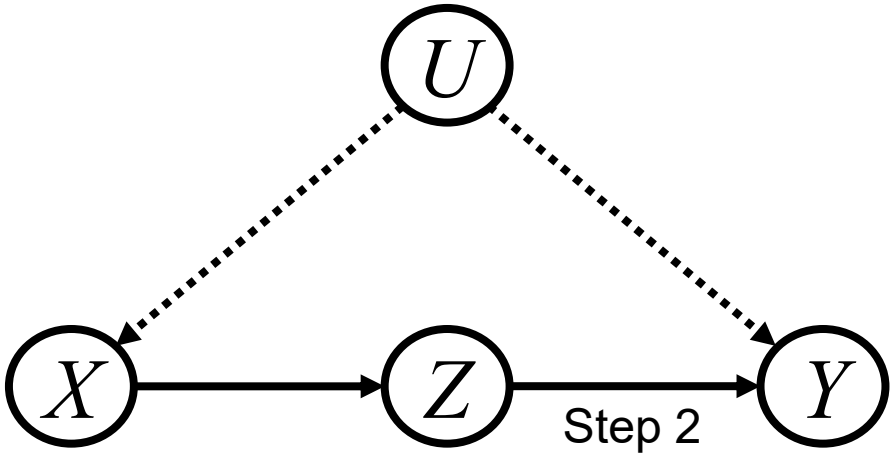
$$P(z|do(x)) = P(z|x)$$



Frontdoor Adjustment: Step 2

- Identify the causal effect of Z on Y

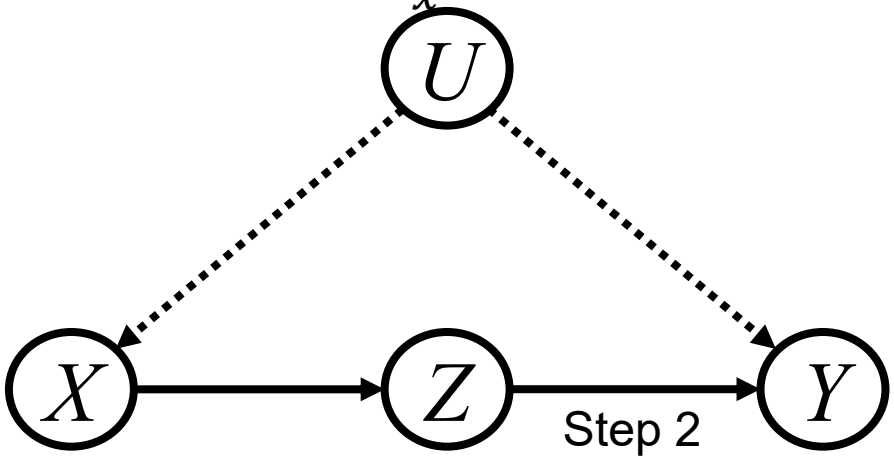
$$P(y|do(z))$$



Frontdoor Adjustment: Step 2

- Identify the causal effect of Z on Y

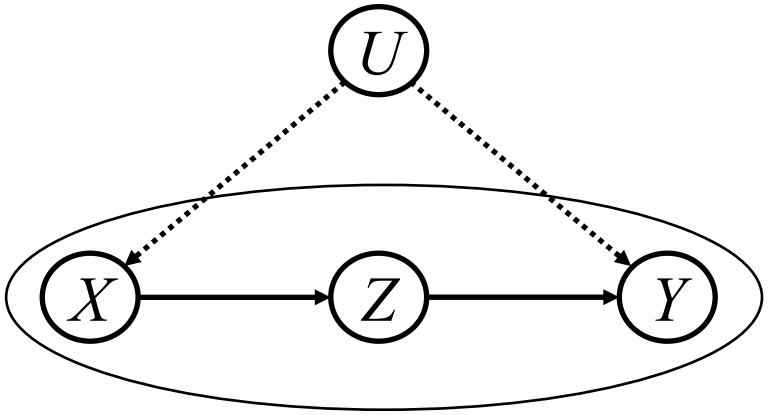
$$P(y|do(z)) = \sum_{x} P(y|z,x)P(x)$$



Frontdoor Adjustment: Step 3

- Combine the above steps to identify the causal effect of X on Y

$$P(y|do(x))$$

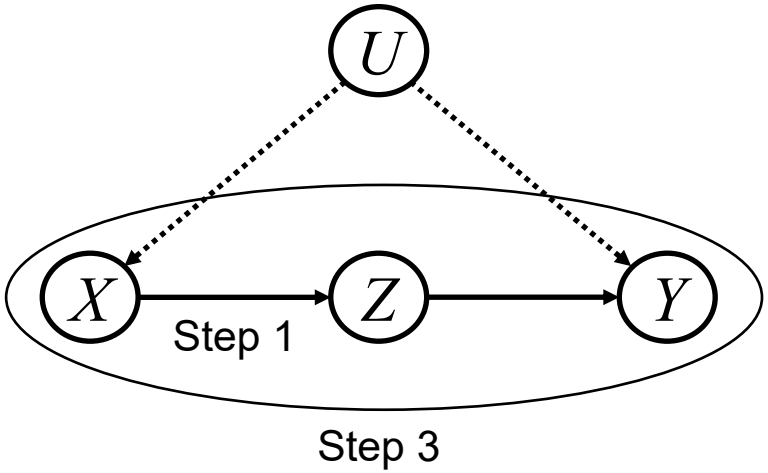


Step 3

Frontdoor Adjustment: Step 3

- Combine the above steps to identify the causal effect of X on Y

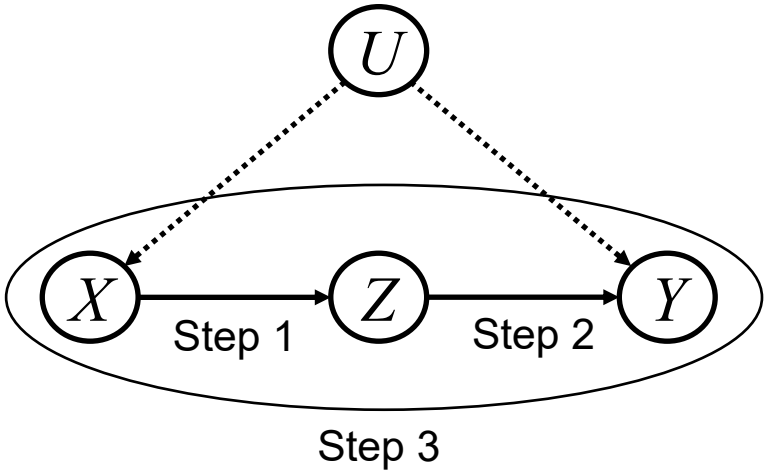
$$P(y|do(x)) \qquad P(z|do(x))$$



Frontdoor Adjustment: Step 3

- Combine the above steps to identify the causal effect of X on Y

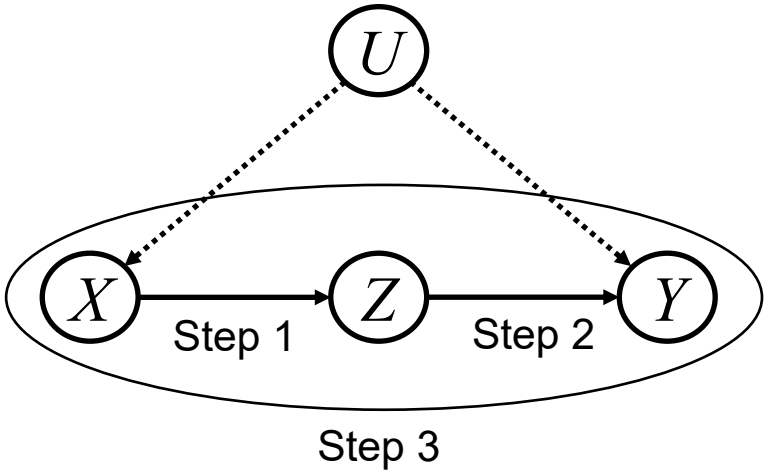
$$P(y|do(x)) \qquad P(z|do(x))P(y|do(z))$$



Frontdoor Adjustment: Step 3

- Combine the above steps to identify the causal effect of X on Y

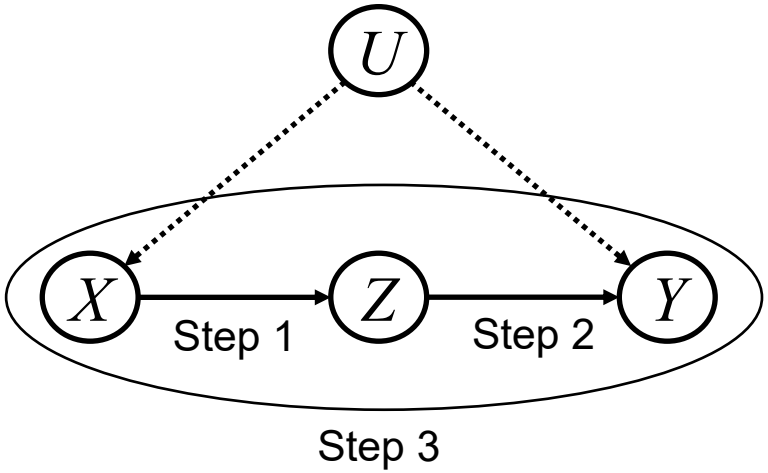
$$P(y|do(x)) = \sum_{z} P(z|do(x))P(y|do(z))$$



Frontdoor Adjustment: Step 3

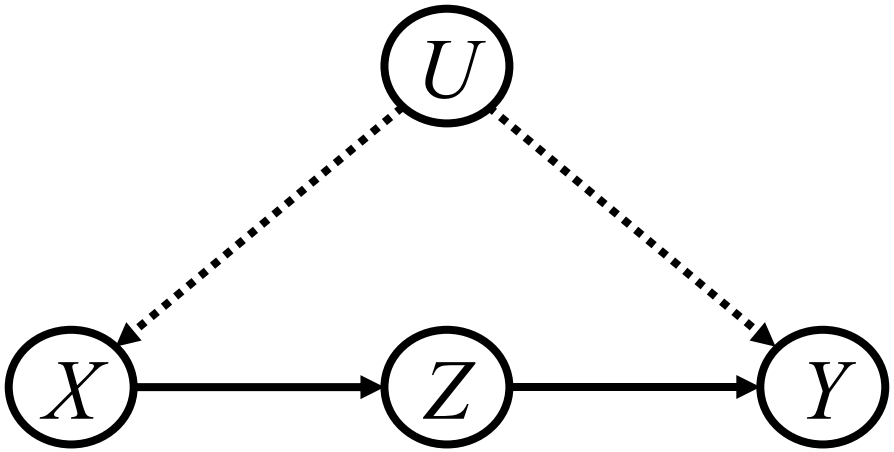
- Combine the above steps to identify the causal effect of X on Y

$$P(y|do(x)) = \sum_{z} P(z|do(x))P(y|do(z))$$
$$= \sum_{z} P(z|x) \sum_{x'} P(y|x', z)P(x')$$



Frontdoor Criterion

- A set of variables Z satisfies the ***frontdoor criterion*** relative to an ordered pair of variables (X, Y) in a DAG G if
  - Z intercepts all directed paths from X to Y;
  - there is no unblocked backdoor path from X to Z; and
  - all backdoor paths from Z to Y are blocked by X.



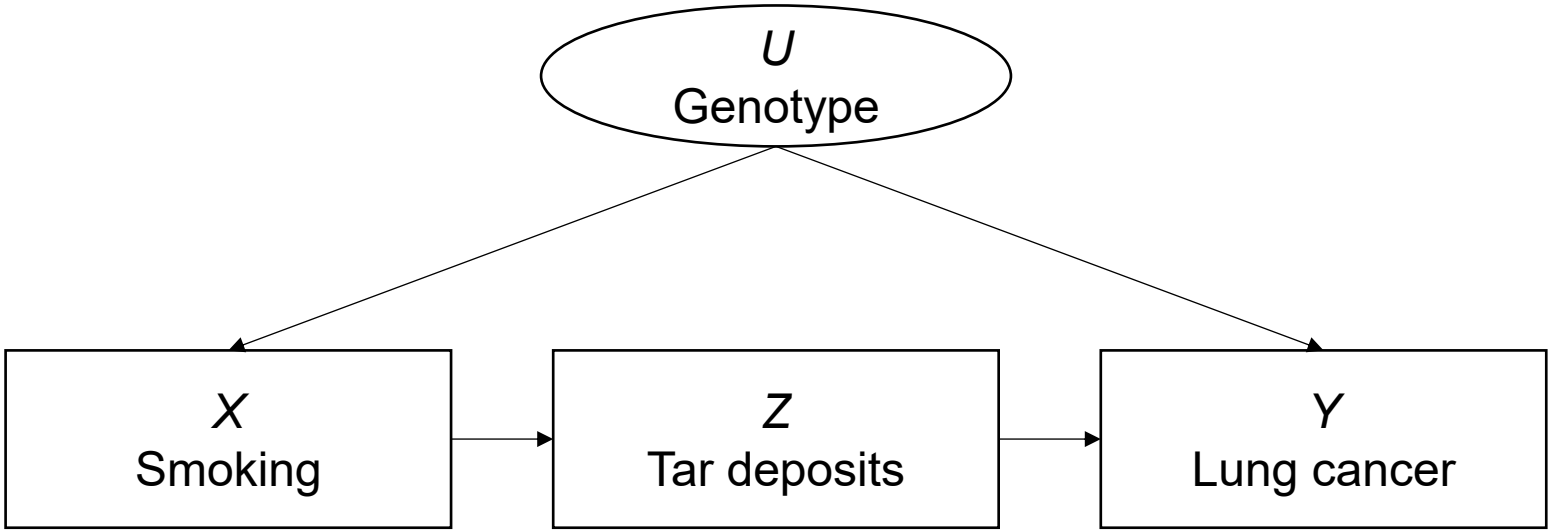
Frontdoor Adjustment

- If a set of variables Z satisfies the ***frontdoor criterion*** relative to (X, Y) and if $P(x, z) > 0$, then the causal effect of X on Y is identifiable and is given by the ***frontdoor adjustment*** formula

$$P(y|do(x)) = \sum_{z} P(z|x) \sum_{x'} P(y|x', z) P(x')$$

Example: Smoking and the Genotype Theory

- According to genotype theory, the observed correlation between smoking(X) and lung cancer(Y) could be explained by some sort of carcinogenic genotype(U) that involves inborn craving for nicotine.



Example: Smoking and the Genotype Theory

- A hypothetical data set

	Group Type	$P(x, z)$ Group Size (% of Population)	$P(Y = 1 \mid x, z)$ % of Cancer Cases in Group
$X = 0, Z = 0$	Non-Smokers, No-tar	47.5	10
$X = 1, Z = 0$	Smokers, No-tar	2.5	90
$X = 0, Z = 1$	Non-Smokers, Tar	2.5	5
$X = 1, Z = 1$	Smokers, Tar	47.5	85

Example: Smoking and the Genotype Theory

- It seems that smoking ***increases*** one's risk of lung cancer: 95% of smokers and only 5% of non-smokers have developed high levels of tar deposits; moreover, 81% of subjects with tar deposits have developed lung cancer, compared to only 9% among those with no tar deposits.

	Group Type	$P(x, z)$ Group Size (% of Population)	$P(Y = 1 \mid x, z)$ % of Cancer Cases in Group
$X = 0, Z = 0$	Non-Smokers, No-tar	47.5	10
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Example: Smoking and the Genotype Theory

- However, genotype theory tells a different story that smoking actually ***decreases*** one's risk of lung cancer.

	Group Type	$P(x, z)$ Group Size (% of Population)	$P(Y = 1 \mid x, z)$ % of Cancer Cases in Group
$X = 0, Z = 0$	Non-Smokers, No-tar	47.5	10
$X = 1, Z = 0$	Smokers, No-tar	2.5	90
$X = 0, Z = 1$	Non-Smokers, Tar	2.5	5
$X = 1, Z = 1$	Smokers, Tar	47.5	85

Example: Smoking and the Genotype Theory

- If you decide to smoke, then your chances of building tar deposits are 95%, compared to 5% if you decide not to smoke.

	Group Type	$P(x, z)$ Group Size (% of Population)	$P(Y = 1 \mid x, z)$ % of Cancer Cases in Group
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Example: Smoking and the Genotype Theory

- In order to evaluate the effect of tar deposits, we look separately at two groups, smokers and non-smokers. In smokers, tar deposits lower cancer rates from 90% to 85%; in non-smokers, they lower cancer rates from 10% to 5%.

	Group Type	$P(x, z)$ Group Size (% of Population)	$P(Y = 1 \mid x, z)$ % of Cancer Cases in Group
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Example: Smoking and the Genotype Theory

- Thus, regardless of whether I have a natural craving for nicotine, I should be seeking the protective effect of tar deposits in my lungs, and smoking offers a very effective means of acquring those deposits.

Example: Smoking and the Genotype Theory

- We now apply the frondoor adjustment formula to the hypothetical data set to prove that smoking actually ***decreases*** one's risk of lung cancer.

	Group Type	$P(x, z)$ Group Size (% of Population)	$P(Y = 1 \mid x, z)$ % of Cancer Cases in Group
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$X = 0, Z = 1$	Non-Smokers, Tar	2.5	5
$X = 1, Z = 1$	Smokers, Tar	47.5	85

Example: Smoking and the Genotype Theory

- $P(Y = 1|do(X = 0))$
- $= P(Z = 0|X = 0)[P(Y = 1|X = 0, Z = 0)P(X = 0) + P(Y = 1|X = 1, Z = 0)P(X = 1)] + P(Z = 1|X = 0)[P(Y = 1|X = 0, Z = 1)P(X = 0) + P(Y = 1|X = 1, Z = 1)P(X = 1)]$
- $= .95(.10 \times .50 + .90 \times .50) + .05(.05 \times .50 + .85 \times .50)$
- $= .4975$

	Group Type	$P(x, z)$ Group Size (% of Population)	$P(Y = 1\|x, z)$ % of Cancer Cases in Group
$X = 0, Z = 0$	Non-Smokers, No-tar	47.5	10
$X = 1, Z = 0$	Smokers, No-tar	2.5	90
$X = 0, Z = 1$	Non-Smokers, Tar	2.5	5
$X = 1, Z = 1$	Smokers, Tar	47.5	85

Example: Smoking and the Genotype Theory

- $P(Y = 1|do(X = 1))$
- $= P(Z = 0|X = 1)[P(Y = 1|X = 0, Z = 0)P(X = 0) + P(Y = 1|X = 1, Z = 0)P(X = 1)] + P(Z = 1|X = 1)[P(Y = 1|X = 0, Z = 1)P(X = 0) + P(Y = 1|X = 1, Z = 1)P(X = 1)]$
- $= .05(.10 \times .50 + .90 \times .50) + .95(.05 \times .50 + .85 \times .50)$
- $= .4525$

	Group Type	$P(x, z)$ Group Size (% of Population)	$P(Y = 1\|x, z)$ % of Cancer Cases in Group
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Example: Smoking and the Genotype Theory

- $P(Y = 1|do(X = 1)) < P(Y = 1|do(X = 0))$. Thus, contrary to naive expectations, the data proves smoking to be beneficial to one's health.

Outline

- Frontdoor adjustment
- Representative works

Deconfounded image captioning: A causal retrospect†

- Causal graph

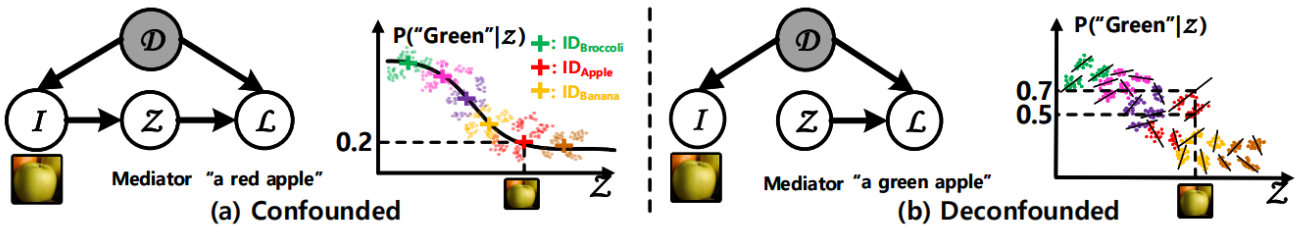


Fig. 3: The front-door IC whose captioning process is $\mathcal{I} \rightarrow \mathcal{Z} \rightarrow \mathcal{L}$, where $\mathcal{Z}$ groups representations into smaller clusters. (a) If this IC is trained by the passive correlation $P(\mathcal{L}|\mathcal{I})$ (Eq. (3)), $\mathcal{Z}$ is still implicitly affected by $\mathcal{D}$ through the path $\mathcal{D} \rightarrow \mathcal{I} \rightarrow \mathcal{Z}$. Then the learnt $P(\text{"}Green\text{"}|\mathcal{Z})$ still crosses the collapsed semantic points as in Fig. 1(b). (b) When we train this IC by the active intervention $P(\mathcal{L}|do(\mathcal{I}))$ (Eq. (4)), the link $\mathcal{I} \rightarrow \mathcal{Z}$ is cut off and this IC is deconfounded.

- Frontdoor adjustment

$$P(\mathcal{L}|do(\mathcal{I})) = \sum_{\boldsymbol{z}} P(\boldsymbol{z}|\mathcal{I}) \sum_{\boldsymbol{x}} P(\mathcal{L}|\boldsymbol{z}, \boldsymbol{x}) p(\boldsymbol{x}).$$

†Yang X, Zhang H, Cai J. Deconfounded image captioning: A causal retrospect[J]. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2021.

Deconfounded image captioning: A causal retrospect

- The intervention distribution after applying backdoor adjustment and frontdoor adjustment is

$$P(\mathcal{L}|do(\mathcal{I})) = \sum\nolimits_{\boldsymbol{s}} P(\boldsymbol{s}) \sum\nolimits_{\boldsymbol{x}} P(\boldsymbol{x}) \sum\nolimits_{\boldsymbol{z}} P(\boldsymbol{z}|\mathcal{I})[P(\mathcal{L}|\boldsymbol{s},\boldsymbol{x},\boldsymbol{z})]$$
$$= \mathbb{E}_{\boldsymbol{s}}\mathbb{E}_{\boldsymbol{x}}\mathbb{E}_{[\boldsymbol{z}|\mathcal{I}]}[P(\mathcal{L}|\boldsymbol{s},\boldsymbol{x},\boldsymbol{z})],$$

which is the expected value of $P(\mathcal{L}|\boldsymbol{s},\boldsymbol{x},\boldsymbol{z})$ according to three variables $\boldsymbol{s}$, $\boldsymbol{x}$, and $\boldsymbol{z}$, which denote the word embeddings of key words, the visual features of the image $\mathcal{I}$, and the embeddings of the semantic structures, respectively.

- Applying Normalized Weighted Geometric Mean(***NWGM***) to absorb the expectations into the network

$$P(\mathcal{L}|do(\mathcal{I})) \approx \mathrm{Softmax}\{g(\mathbb{E}_{\boldsymbol{s}}[\boldsymbol{s}], \mathbb{E}_{\boldsymbol{x}}[\boldsymbol{x}], \mathbb{E}_{[\boldsymbol{z}|\mathcal{I}]}[\boldsymbol{z}])\}$$

Deconfounded image captioning: A causal retrospect

- Sketch of DIC(Deconfouned Image Captioning)

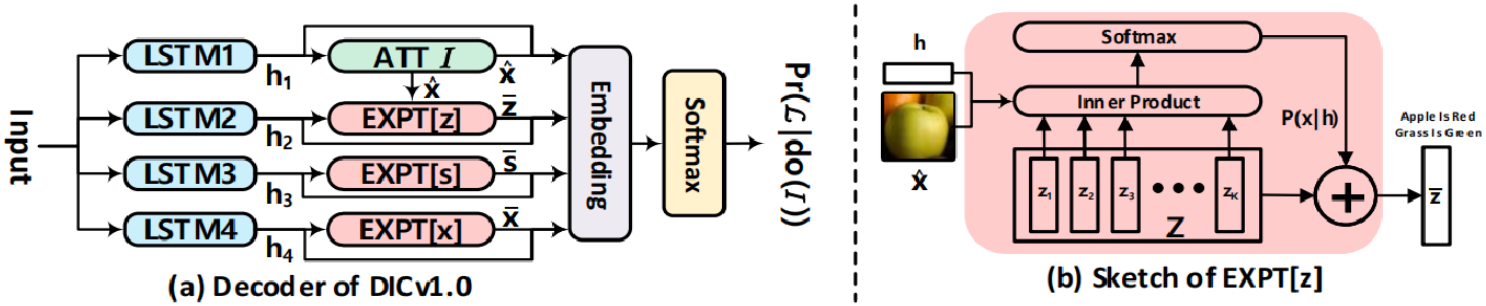


Fig. 5: (a) The sketch of our DICv1.0's decoder, where ATT/EXPT represent attention/expectation modules. (b) The sketch of EXPT[z]

Causal attention for vision-language tasks†

- Causal graph

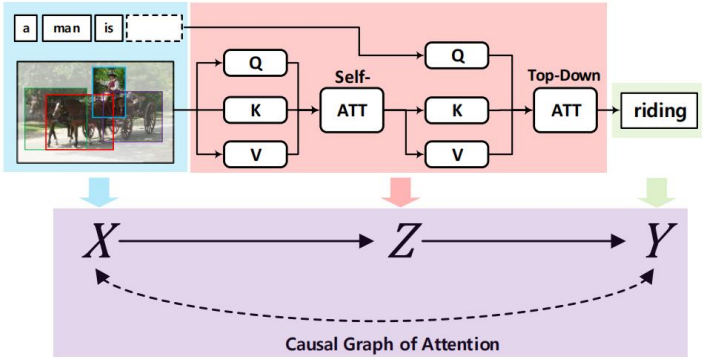


Figure 1. Top: an example of image captioner with a self-attention and a top-down attention modules. Bottom: the corresponding causal graph.

- Frontdoor adjustment

$$P(Y|do(X)) = \underbrace{\sum_z P(Z=z|X)}_{\text{IS-Sampling}} \underbrace{\sum_x P(X=x)}_{\text{CS-Sampling}} [P(Y|Z=z, X=x)].$$

†Yang X, Zhang H, Qi G, et al. Causal attention for vision-language tasks[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2021: 9847-9857.

Causal attention for vision-language tasks

- Specifically, by ***NWGM*** approximation, IS-Sampling and CS-Sampling in can be absorbed into the network as follows

$$P(Y|do(X)) \approx \text{Softmax}[g(\hat{\boldsymbol{Z}}, \hat{\boldsymbol{X}})],$$

g is a network

$$\textbf{IS-Sampling:} \quad \hat{\boldsymbol{Z}} = \sum\nolimits_z P(Z = z|h(X))\boldsymbol{z},$$

$$\textbf{CS-Sampling:} \quad \hat{\boldsymbol{X}} = \sum\nolimits_x P(X = x|f(X))\boldsymbol{x}.$$

where $h(\cdot)$ and $f(\cdot)$ denote query embedding functions which can transform the input X into two query sets.

Causal attention for vision-language tasks

- Illustration for causal attention

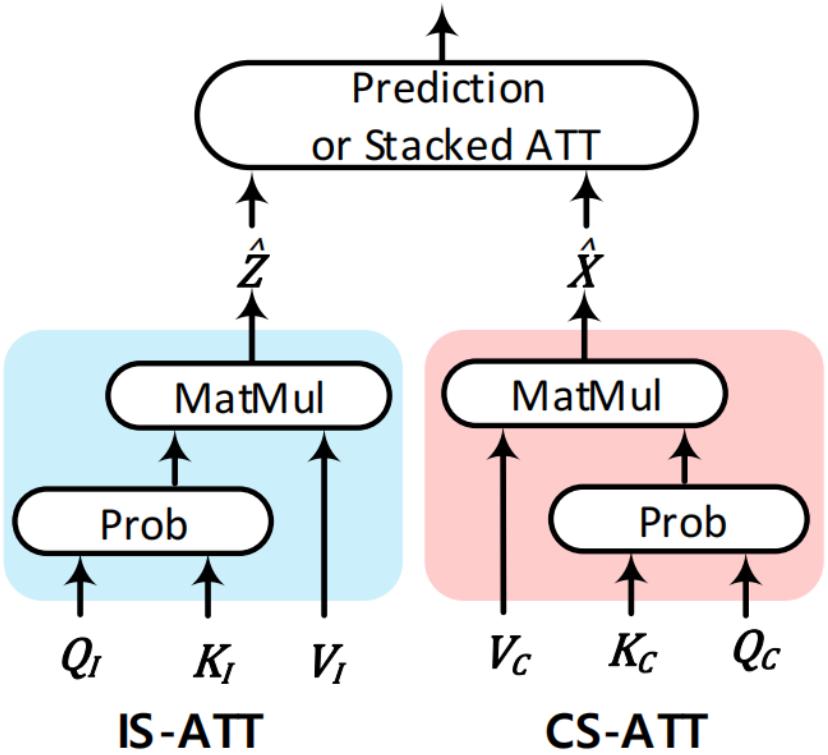


Figure 4. The sketch of a single causal attention module, which includes an IS-ATT (Eq. (6)) and a CS-ATT (Eq. (7)). After calculating $\hat{Z}$ and $\hat{X}$, we can input them into the predictor for making decisions or more stacked attention layers for further embedding.

Causal Transportability for Visual Recognition†

- Causal graph

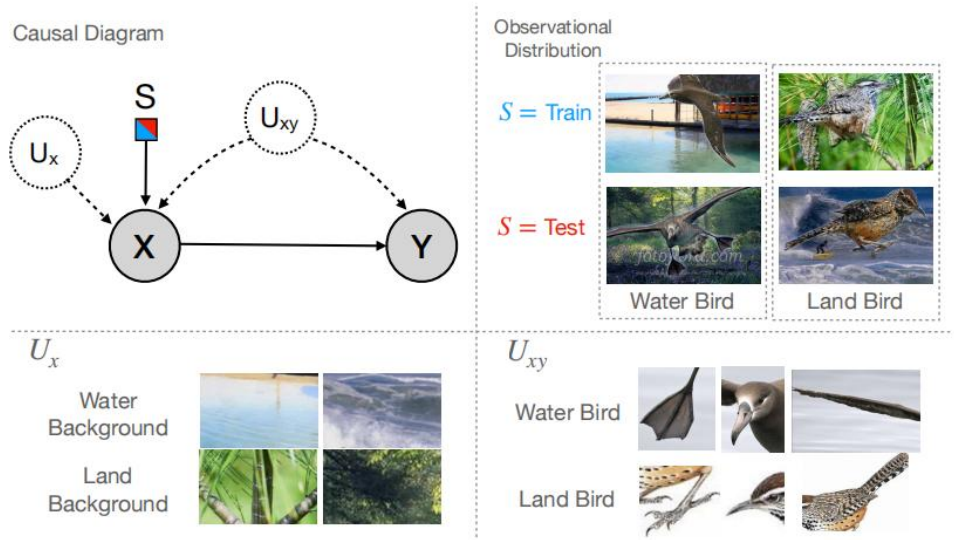


Figure 1. Causal graph for out-of-distribution image classification (top left). Image X is constructed from nuisance features U_X (bottom left) and concept features U_{XY} (bottom right). Label Y is created from X and U_{XY}. S, the transportability node, points to nodes with changes between domains, where X combines 'waterbird' with 'water background' during the training ($S = 0$) and 'water bird' with 'land background' at testing ($S = 1$) (top right).

†Mao C, Xia K, Wang J, et al. Causal Transportability for Visual Recognition[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2022: 7521-7531.

Causal Transportability for Visual Recognition

- Expanded causal graph

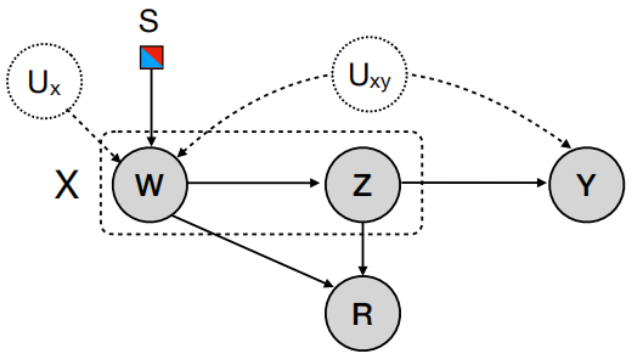


Figure 3. Expanded causal model with decomposition of image X and representation R. Gray nodes denote observed variables.

- Frontdoor adjustment

Theorem 1 (Causal Identification). *Given the assumptions about the generative process encoded in the causal graph in Fig. 3 together with assumptions 1, 2, 3, the causal effect can be computed using neural representation R via* $P(Y = y|do(X = x)) = \sum_r \hat{P}(r|x) \sum_{x'} \hat{P}(y|r, x')P(x')$.

Causal Transportability for Visual Recognition

- Training algorithm

Algorithm 1 Causal-Transportability Model Training

Input: Training set D over $\{(X, Y)\}$.
Phase 1: Compute $\hat{P}(R|X)$ from representation of VAE or pretrained model.
Phase 2:
for $i = 1, ..., K$ **do**
  Sample x_i, r_i, y_i from the joint distribution $D' = (X, R, Y)$
  Random sample x_i' from the same category as x_i
  Train $\hat{P}(Y|X', R)$ via minimizing the classification loss $\mathcal{L}$ through gradient descent.
end for
Output: Model $\hat{P}(R|X)$ and $\hat{P}(Y|X, R)$

Causal Transportability for Visual Recognition

- Testing algorithm

Algorithm 2 Causal-Transportability Effect Evaluation

Input: Query x, training distribution D over $\{(X, Y)\}$, model $\hat{P}(R|X)$ and $\hat{P}(Y|X', R)$, the sampling time N_i for the representation variable R, and the sampling time N_j for X'.
for $i = 1, ..., N_i$ **do**
  $\mathbf{r}_i \leftarrow \hat{P}(r|x)$
  for $j = 1, ..., N_j$ **do**
    Random sample $\mathbf{x}'_{ij}$ from Training Distribution D.
    Compute $\hat{P}(Y|x'_{ij}, r_i)$
  end for
end for
Calculate the causal effect $P(y|\text{do}(X = x)) = \sum_i \hat{P}(r_i|x) \sum_j \hat{P}(y|r_i, x'_{ij}) P(x'_{ij})$
Output: Class $\hat{y} = \text{argmax}_y P(y|\text{do}(X = x))$.

The Role of Deconfounding in Meta-learning†

- Model-Agonstic Meta-Learning(MAML)
  - Inner loop

$$\mathcal{L}(\phi_i) = \frac{1}{K^s} \sum_{j=1}^{K^s} \mathcal{L}(f_{\phi_i,\theta}(x_{i,j}^s), y_{i,j}^s),$$

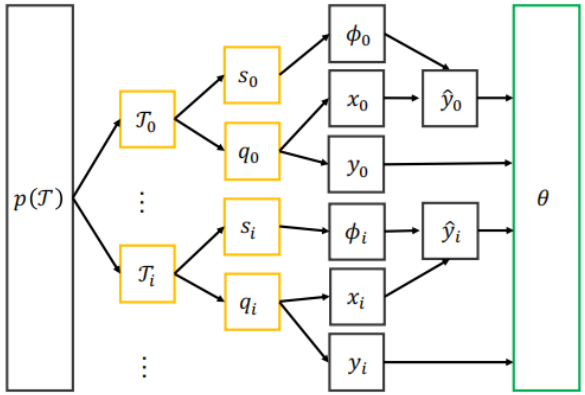
  - Outer loop

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}_{p(\phi_i|\theta,s_i)}[\frac{1}{K_i^q} \sum_{j=1}^{K_i^q} \mathcal{L}(f_{\phi_i,\theta}(x_{i,j}^q), y_{i,j}^q)].$$

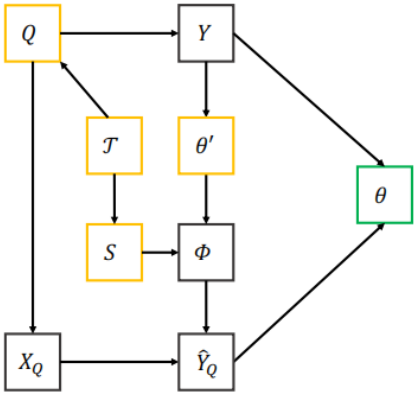
†Jiang Y, Chen Z, Kuang K, et al. The Role of Deconfounding in Meta-learning[C]//International Conference on Machine Learning. PMLR, 2022: 10161-10176.

The Role of Deconfounding in Meta-learning

- Workflow and causal graph



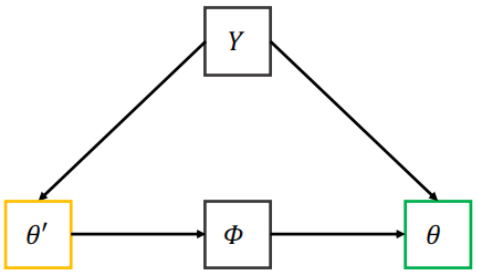
(a) Workflow of meta-learning



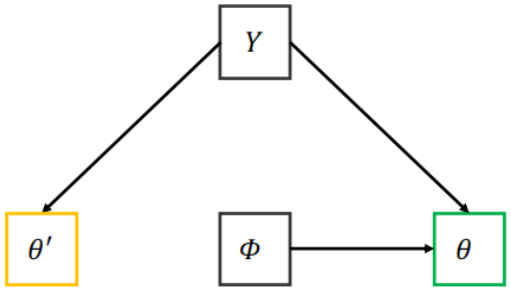
(b) Causal graph of meta-learning

The Role of Deconfounding in Meta-learning

- Simplified causal graph



- Deconfouding MAML



(d) Deconfounded by front-door adjustment.

The Role of Deconfounding in Meta-learning

- Frontdoor adjustment

$$
\begin{aligned}
p(\theta|do(\theta'), S, Q) &= \sum_{\Phi} p(\Phi|\theta', S) p(\theta|do(\Phi), Q) \\
&= \sum_{\Phi} p(\Phi|\theta', S) \sum_{\theta'_i} p(\theta|\Phi, \theta'_i, Q) p(\theta'_i) \\
&= \sum_{\theta'_i} p(\theta|\Phi, \theta'_i, Q) p(\theta'_i),
\end{aligned}
\tag{2}
$$

where $p(\Phi|\theta', S) = 1$ denotes the delta function.

The Role of Deconfounding in Meta-learning

- MAML-Dropout

$$
\begin{aligned}
p(\theta|do(\Phi), Q) &= \int p(\theta|\Phi, \theta'_i, Q) p(\theta'_i) d\theta'_i \\
&\approx \frac{1}{M} \sum_{i=1}^{M} p(\theta|\Phi, \theta'_i, Q) \\
&= \frac{1}{M} \sum_{i=1}^{M} p(\theta|\Phi, \theta', Q, z_i),
\end{aligned}
$$

- MAML-Bins

$$
p(\theta|do(\Phi), Q) = \frac{1}{M} \sum_{M} p(\theta|\Phi, \theta'_m, Q).
$$

Causal Class Activation Maps for Weakly-Supervised Semantic Segmentation†

- Causal graph

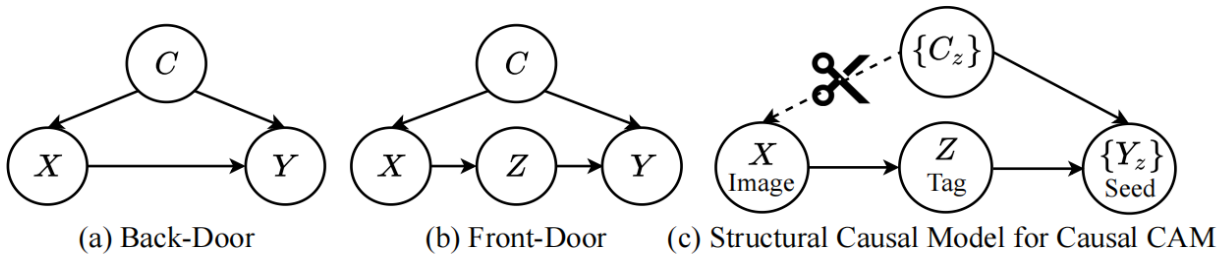


Figure 1: Structural Causal Models (SCMs) for illustrating the fundamental (a) Back-Door model and (b) Front-Door model (c) presents our proposed SCM to analyze the generation of the localization Y_z, in which the image X, image-level tag Z, pixel-level localization Y_z, and class-specific confounder C_z can be formulated in a front-door model. The "scissor" in (c) denotes causal intervention.

†Wang Y. Causal Class Activation Maps for Weakly-Supervised Semantic Segmentation[C]//UAI 2022 Workshop on Causal Representation Learning.

Causal Class Activation Maps for Weakly-Supervised Semantic Segmentation

- Frontdoor adjustment

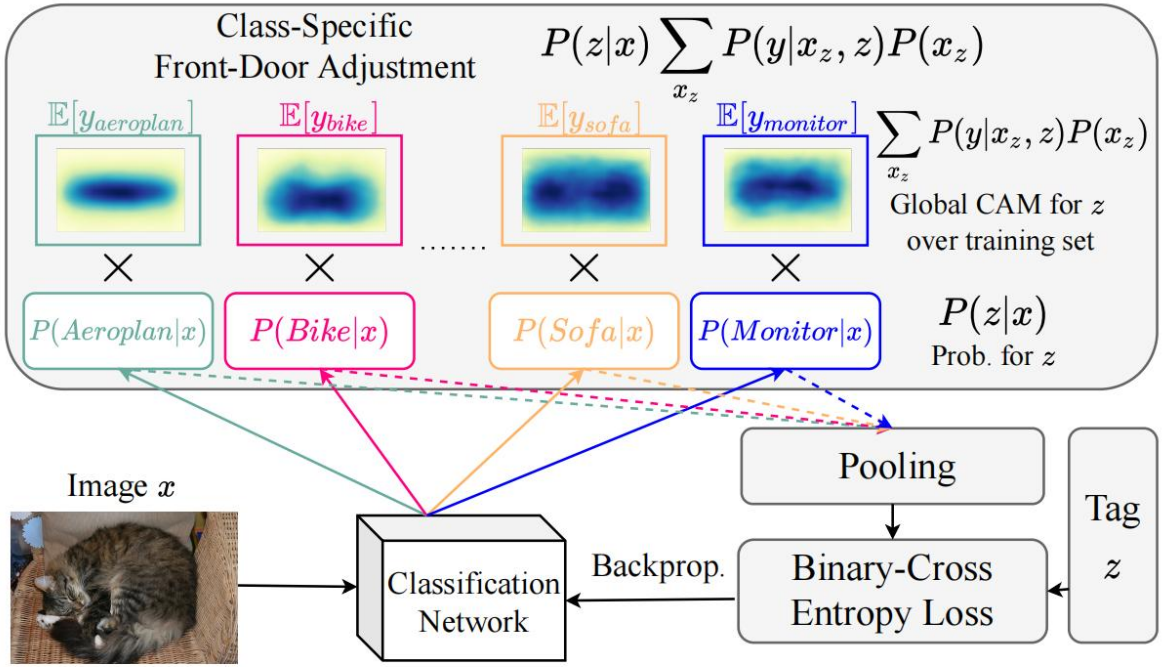
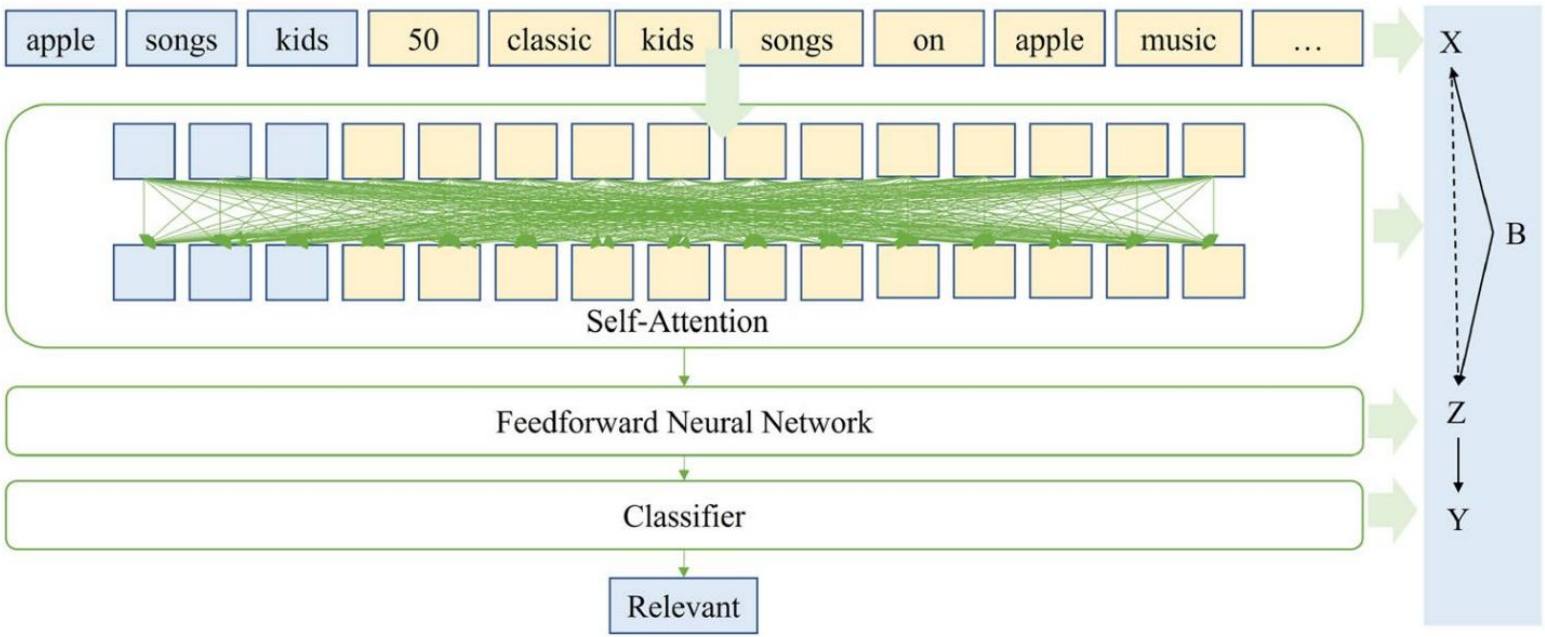


Figure 2: Overview of the proposed method of applying front-door adjustment for a classification network.

Disentangled Graph Recurrent Network for Document Ranking†

- Causal graph



†Dong Q, Niu S, Yuan T, et al. Disentangled Graph Recurrent Network for Document Ranking[J]. Data Science and Engineering, 2022, 7(1): 30-43.

Disentangled Graph Recurrent Network for Document Ranking

- Frontdoor adjustment

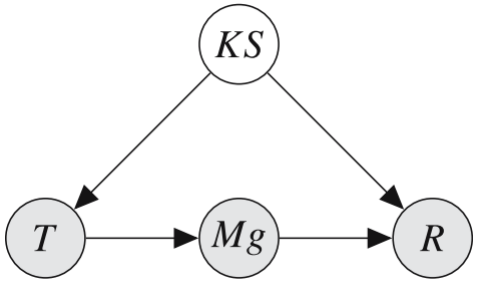
$$P(Y|do(X)) = \sum_{z} P(Y|do(Z))P(Z|do(X))$$

- Appyling Weighted Geometric Mean

$$P(Y|do(X)) = \mathbb{E}_Z \mathbb{E}_B[P(Y|X,Z)]$$
$$\propto \exp\left(g(\mathbb{E}_Z[Z], \mathbb{E}_B[X])\right)$$

Estimating causal effects with the neural autoregressive density estimator†

- The kidney stone example contains three variables: stone size (*KS*), treatment type(*T*), and Recovery (*R*).
- Causal graph



- Frontdoor adjustment

$$P(R|\mathrm{do}(T=\hat{t})) = \int_{mg} P(\mathrm{Mg}=mg|T=\hat{t}) \int_{t'} P(R|\mathrm{Mg}=mg, T=t')P(T=t').$$

†Garrido S, Borysov S, Rich J, et al. Estimating causal effects with the neural autoregressive density estimator[J]. Journal of Causal Inference, 2021, 9(1): 211-218.

Evaluating the Causal Effect of Multimedia and Affective Temperament in Felt Emotion and Liking†

- Variables description and causal graph

Table 1. Variables description

Name	Data Type	Description
Affective Temperament	Integer (1 to 4)	The result of the AFECT questionnaire [7] for each volunteer, the numbers mean the 4 categories mentioned on subsection 2.2.
Gender	Integer (0 or 1)	The gender of the participant, 0 being male and 1 being female
EEG	Integer (1 to 4)	Five HOC features were extracted from 4 EEG channels (FP1, FP2, F3, F4, *i.e.*, 20 levels per video), PCA was computed from the features to reduce them to 1 dimension (explained variance ratio of 68,26%). The PCA values of the videos were discretized by cutting the array in 4 same-length ranges: the bins were named 1, 2, 3 and 4.
Video Concept	Integer (0 to 3)	Each video was attributed a cluster of concepts by the k-means algorithm (4 groups)
AV-Quadrant	Integer (1 to 4)	Arousal-Valence plane quadrant (Fig.6) attributed to each video by each participant right after watching it (the participants informed the emotion felt in a 9-point scale for valence, likewise for arousal)
Liking	Integer (1 to 3)	Each participant informed their liking about each video in a 3-point scale (disliked, neutral, liked)

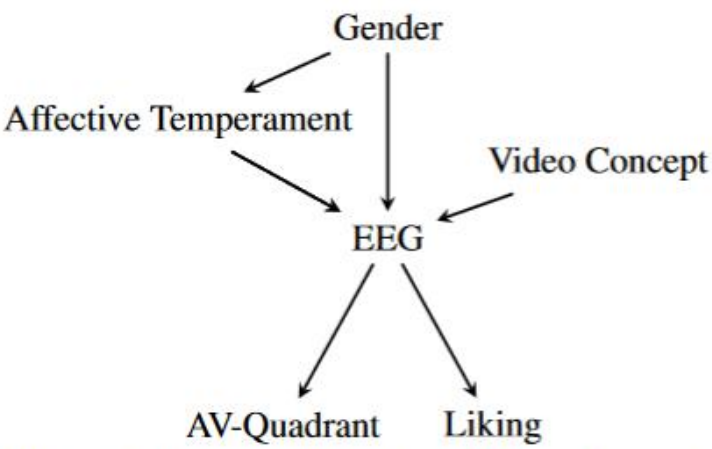


Figure 8. Proposed hypothetical Causal Graph.

†Silva R A B, Pereira E T. Evaluating the Causal Effect of Multimedia and Affective Temperament in Felt Emotion and Liking[J]. Revista de Informática Teórica e Aplicada, 2022, 29(2): 35-47.

Evaluating the Causal Effect of Multimedia and Affective Temperament in Felt Emotion and Liking

- Frontdoor adjustment

Table 4. Causal Effect of Gender on Liking (Eq. 9)

	Liking		
Gender	1	2	3
0	0.3583	0.3310	0.3107
1	0.3649	0.3336	0.3016

$$P(Liking|do(Gender)) = \sum_{EEG} P(EEG|Gender) \cdot \sum_{Gender'} P(Liking|Gender', EEG) \cdot P(Gender') \quad (9)$$

Confounder Identification-free Causal Visual Feature Learning†

- Causal graph

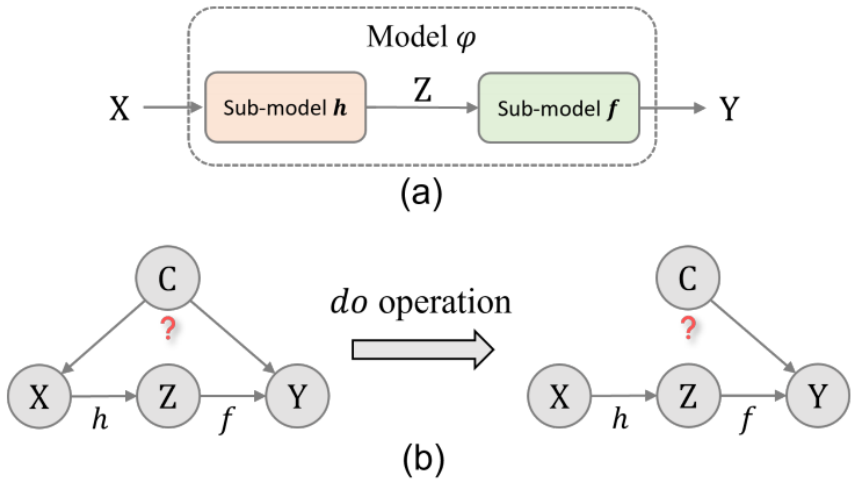


Fig. 2. Causal graph in supervised learning. (a) A network/model φ consists of sub-models h and f. X and Y denote the input and label, respectively. (b) Causal graph where *do* operation aims to remove the intervening from confounder C.

†Li X, Zhang Z, Wei G, et al. Confounder Identification-free Causal Visual Feature Learning[J]. arXiv preprint arXiv:2111.13420, 2021.

Confounder Identification-free Causal Visual Feature Learning

- Frontdoor adjustment

$$P(Y|do(X=x)) = \sum_{\tilde{x} \in X} P(Y|Z=h(x),\tilde{x})P(\tilde{x}),$$

- A gradient-based instantiation

$$P(Y|Z=h(x),\tilde{x}) = f_{\theta_{\tilde{x}}}(Z=h(x)),$$
$$where\ \theta_{\tilde{x}} = \theta - \alpha g_{\tilde{x}},\ \ g_{\tilde{x}} = \nabla_{\theta}\mathcal{L}(f_{\theta}(h(\tilde{x}),\tilde{y})),$$

Confounder Identification-free Causal Visual Feature Learning

- Algorithm

Algorithm 1 Confounder Identification-free Causal Visual Feature Learning

Input: Training dataset $\{(x_i, y_i)\}_{i=1}^{N}$.
Init: learning rate: α, β; model f with parameter θ.
Cluster the training data into $K^{\dagger}$ clusters. ▷ Figure 3
while not converge **do**
    Sample M samples from $K^{\dagger}$ clusters as a batch.
    Estimate global intervention with $g_{\dagger}$. ▷ Eq. 8
    Update f with $g_{\dagger}$ as: $\theta_{\dagger} = \theta - \alpha g_{\dagger}$.
    Compute the loss $\mathcal{L}_{CICF}$. ▷ Eq. 11
    Update $\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}_{CICF}$.
end while

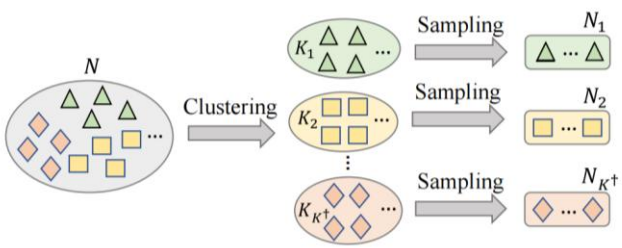
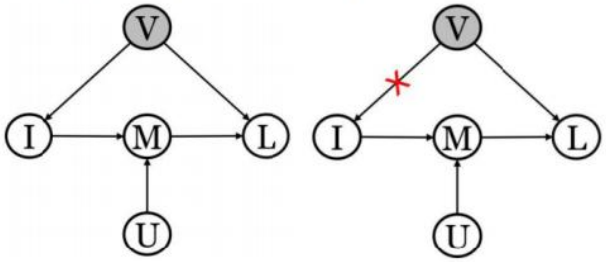


Fig. 3. Clustering-then-sampling strategy, where N, K_k, and N_k denote the number of samples, the kth cluster, and the number of sampled samples from the kth cluster, respectively.

Mitigating Hidden Confounding Effects for Causal Recommendation†

- Abstract user feedback generation process into causal graph.
  - V: hidden confounder; L: like feedback; I: item; U: user.
  - M: a set of variables that act as mediators between $\{U, I\}$ and L, e.g., user-item feature matching, and click.
- Key:
  - Block the backdoor path $I \leftarrow V \rightarrow L$
  - Estimate the causal effect of I on L, *i.e.*,

$$P(L|U, \mathrm{do}(I)).$$



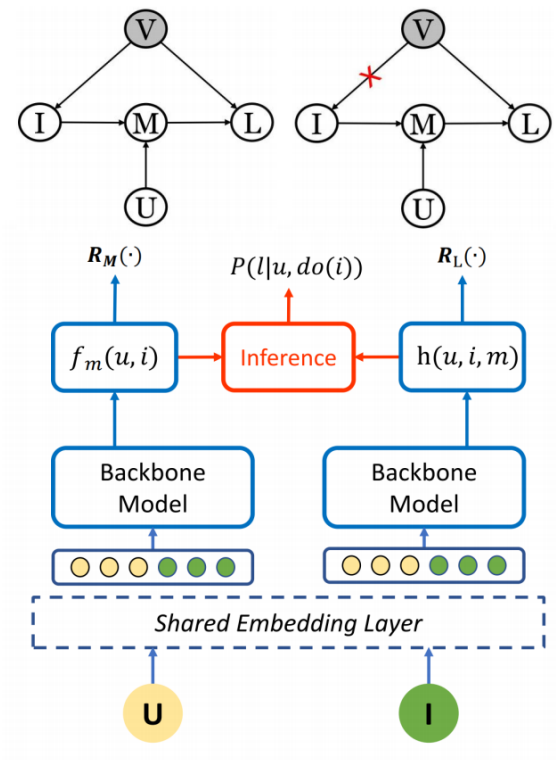
- Hidden Confounder Removal (HCR) framework.
  - Front-door adjustment
    - decompose causal effect of I on L into: 1) the effects of I on M and 2) the effect of M on L.

$$P(L|U, \mathrm{do}(I)) = \sum_M P(M|U, do(I)) P(L|U, do(M))$$
$$= \sum_M P(M|U, I) \sum_{I'} P(I') P(L|M, U, I')$$

†Zhu X, Zhang Y, Feng F, et al. Mitigating Hidden Confounding Effects for Causal Recommendation[J]. arXiv preprint arXiv:2205.07499, 2022.

Mitigating Hidden Confounding Effects for Causal Recommendation

- Hidden Confounder Removal (HCR) framework
  - $P(L|do(I),U) = \sum_M P\ (M|U,I) \sum_{I'} P\ (I')P\ (L|U,I',M)$
  - Multi-task learning
    - Learns $P(M|U,I) \coloneqq f_m(U,I)$
    - Learn
      $$\mathrm{P}(L|M,U,I) \coloneqq h(U,I,M) = h^1(U,M)h^2(U,I')$$
  - Inference
    - Infer $P(M|U,I)$ and $P(L|U,I,M)$
    - Get rid of the sum over I and obtain
      $$P(L|U,do(I)) = \sum_M f_m(U,I) \sum_{I'} P(I')h^1(U,M)h^2(U,I') = \sum_M f_m(U,I)h^1(U,M) \sum_{I'} P(I')h^2(U,I') = S_u \sum_M f_m(U,I)h^1(U,M)$$



Summary

- When confounder is unobservable, we can use frontdoor adjustment to identify causal effect with the help of mediator.
- Frontdoor adjustment can be applied to various domains like statistics, machine learning, and recommendation system.